

A Full Integrated Report for the Oncology PK/PD and Simulation-Based Inference Curriculum

Notebook-by-Notebook Theory, Outputs, and Identifiability Lessons

Technical Report

May 20, 2026

Abstract

This report expands the original write-up for `00_clinical_audit_and_model_discovery.ipynb` into a full technical document covering the complete notebook sequence through Notebook 06B. The project starts with heterogeneous longitudinal oncology trial files, builds a canonical tumor-size and dose layer, fits mechanistic tumor-growth inhibition models, studies their hold-out and rolling-origin forecasting behavior, calibrates predictive uncertainty, selects clinically coherent simulation-based inference (SBI) starter strata, constructs real-template synthetic simulators, and then tests neural posterior estimation (NPE) on anti-angiogenic synthetic data. The central result is not a success story about increasingly complex neural models. The central result is an identifiability diagnosis: sparse relative SLD trajectories can support useful empirical forecasting and modest posterior predictive uncertainty quantification, but they weakly identify detailed mechanistic PK/PD parameters even under simulator-matched synthetic data. Consequently, the correct next milestone is posterior predictive validation and uncertainty calibration, not raw parameter recovery on real patients.

Contents

1 High-Level Problem Statement

We observe cancer patients indexed by

$$i = 1, \dots, N.$$

Patient i receives one or more anticancer drugs over time. Let the dose history be

$$u_i = \{(t_{ik}^{\text{dose}}, d_{ik}, a_{ik})\}_{k=1}^{K_i},$$

where t_{ik}^{dose} is the administration time, d_{ik} is the amount, and a_{ik} identifies the drug or regimen.

Tumor burden is observed at scan times

$$t_{ij}^{\text{scan}}, \quad j = 1, \dots, T_i,$$

through an SLD-like scalar $y_{ij} > 0$. The relative tumor trajectory is

$$r_{ij} = \frac{y_{ij}}{y_{i1}},$$

and on log scale

$$\ell_{ij} = \log r_{ij} = \log y_{ij} - \log y_{i1}.$$

The scientific target is a patient-specific latent parameter vector

$$\theta_i \in \Theta \subset \mathbb{R}^p$$

containing quantities such as growth rates, transit rates, treatment effect sizes, resistance fractions, or carrying-capacity parameters. The eventual Bayesian inverse problem is

$$p(\theta_i | x_i),$$

where x_i encodes scan times, relative tumor sizes, dose history, and covariates.

This report explains why the project did *not* jump directly to SBI. The full workflow was

clinical audit $\rightarrow$ mechanistic fitting $\rightarrow$ hold-out validation $\rightarrow$ rolling-origin baselines $\rightarrow$ probabilistic calibration

rather than

raw longitudinal data $\rightarrow$ neural posterior estimation immediately.

The logic is simple: if the simulator family is structurally inadequate, or if the data weakly identify its parameters, a learned posterior can look sophisticated while remaining scientifically misleading.

2 Global Mathematical Framing

2.1 Observation and Loss Geometry

Throughout the notebook sequence, the main residual is defined in log-relative tumor space:

$$e_j(\theta) = \log r_j - \log \hat{r}(t_j; \theta, u).$$

This is appropriate because multiplicative tumor-size errors are more clinically realistic than additive millimeter errors. A symmetric doubling/halving discrepancy gives equal-magnitude residuals with opposite sign.

The standard log-RMSE is

$$\text{RMSE}_{\log}(\theta) = \sqrt{\frac{1}{T} \sum_{j=1}^T e_j(\theta)^2}.$$

A value near 0.30 corresponds to a typical multiplicative error of

$$\exp(0.30) - 1 \approx 35\%.$$

2.2 Forward-Simulator View

Every notebook after the audit phase is implicitly organized around a forward simulator

$$\theta \sim p(\theta), \quad x \sim p(x | \theta),$$

even when the immediate task is deterministic fitting. The SBI question only becomes meaningful once we know:

1. what the latent dynamics are,
2. what the observation model is,
3. what forecasting baselines the simulator must beat,
4. whether the parameters are even identifiable from the chosen observation channel.

The rest of the report follows the notebook chronology while keeping this inverse-problem perspective explicit.

3 Notebook 00: Clinical Audit and First Model Discovery

Notebook 00 is the foundation of the whole project. It is the notebook originally documented in `notebook1.tex`, so this section preserves its depth and then connects it to the later notebooks.

3.1 Canonical Clinical Data Layer

The raw data tree contains multiple oncology trials with different schemas, file names, and measurement conventions. Each discovered study s is mapped into a canonical object

$$\mathcal{D}_s = (O_s, D_s, C_s),$$

where

- O_s is the canonical tumor observation table,
- D_s is the canonical dose table,
- C_s is the canonical covariate table.

The observation table contains columns such as

<code>study_id,</code>	<code>patient_id,</code>
<code>raw_day,</code>	<code>time_days,</code>
<code>sld_mm,</code>	<code>rel_sld,</code>
<code>baseline_sld_mm,</code>	<code>n_lesions,</code>
<code>arm.</code>	

The relative tumor burden is computed patient-wise as

$$\text{rel_sld}_{ij} = \frac{\text{sld_mm}_{ij}}{\text{baseline_sld_mm}_i}.$$

The dose table stores

`study_id, patient_id, time_days, drug, dose_amount, dose_unit, arm,`

while the covariate table keeps demographic and arm metadata for later hierarchical priors or covariate-conditioned SBI.

3.2 Dataset Discovery and Eligibility

The loader recursively searches for known trial-specific patterns and maps them into the common schema. A patient is considered PK/PD-eligible when the trajectory has enough scans and a plausible baseline near treatment initiation. Typical filters are of the form

$$T_i \geq 3, \quad t_{i1} \in [-60, 14] \text{ days.}$$

Duplicate same-day observations are collapsed, typically via a within-day median,

$$\tilde{y}_i(t) = \text{median}\{y_{ij} : t_{ij} = t\},$$

so that the ODE solver only sees strictly increasing evaluation times.

3.3 Audit Outputs

Notebook 00 loaded ten studies and produced the canonical audit summarized in Table ?? . The table is important because it explains why later notebooks focus on certain strata and not others: scan counts, dose completeness, and tumor-size distributions differ meaningfully across studies.

Table 1: Canonical clinical audit summary from Notebook 00.

Study	Obs. pts.	Eligible pts.	Obs. rows	Dose rows	Median scans	Median baseline SI
A6181122	378	321	2,307	23,046	6.0	115
DATASET_263	945	671	4,139	45,516	4.0	120
DATASET_264	935	804	5,369	92,188	5.0	134
EFC10262	592	446	2,953	30,117	4.0	110
EFC5505	432	358	2,001	24,901	5.0	110
EFC4972	320	273	1,503	23,639	5.0	90
AMGEN_20050203	488	387	2,476	0	4.0	157
AMGEN_20020408	275	69	672	0	2.0	190
CEDIRANIB_HORIZONIII	688	584	3,340	46,174	5.0	103
AMGEN_20040249	809	655	3,802	67,273	4.0	127

3.4 Exposure Approximation

The first mechanistic models do not yet use full pharmacokinetics. Instead, dose events are converted into a simple concentration-like forcing curve

$$C(t) = \sum_{k:t_k \leq t} d_k \exp[-k_{\text{el}}(t - t_k)].$$

Dose amounts are normalized by the median positive dose amount so that k_{kill} can be given broad but stable priors. This simplification is deliberate: the project first tests whether the tumor-size signal alone supports meaningful inverse inference before adding additional PK complexity.

3.5 Baseline Simeoni Tumor-Growth Inhibition Model

The baseline simulator is a four-state Simeoni-style model with proliferating tumor x_1 and three damaged/transit compartments x_2, x_3, x_4 :

$$V(t) = x_1(t) + x_2(t) + x_3(t) + x_4(t).$$

The patient-specific parameter vector is

$$\theta = (\lambda_0, \lambda_1, \psi_g, k_{\text{kill}}, k_{\text{tr}}, v_0).$$

The saturating growth function is

$$g(x_1; \lambda_0, \lambda_1, \psi_g) = \frac{\lambda_0 x_1}{\left[1 + \left(\frac{\lambda_0}{\lambda_1} x_1\right)^{\psi_g}\right]^{1/\psi_g}}.$$

For small x_1 , the dynamics are approximately exponential, $g(x_1) \approx \lambda_0 x_1$. For large x_1 , the growth rate approaches an approximately linear regime, $g(x_1) \approx \lambda_1$.

The ODE system is

$$\begin{aligned}\frac{dx_1}{dt} &= g(x_1) - k_{\text{kill}} C(t) x_1, \\ \frac{dx_2}{dt} &= k_{\text{kill}} C(t) x_1 - k_{\text{tr}} x_2, \\ \frac{dx_3}{dt} &= k_{\text{tr}} (x_2 - x_3), \\ \frac{dx_4}{dt} &= k_{\text{tr}} (x_3 - x_4),\end{aligned}$$

with

$$x(0) = (v_0, 0, 0, 0).$$

The fitted observable is the relative trajectory

$$\hat{r}(t_j; \theta) = \frac{V(t_j; \theta)}{V(t_1; \theta)}.$$

3.6 Optimization and Robust Residuals

The project optimizes on transformed parameters $z = \log \theta$ to enforce positivity and improve numerical conditioning. Both squared-error and Huber-style robust losses are used in log-relative space. For threshold δ ,

$$\rho_\delta(e) = \begin{cases} \frac{1}{2} e^2, & |e| \leq \delta, \\ \delta (|e| - \frac{1}{2} \delta), & |e| > \delta. \end{cases}$$

The corresponding robust aggregate score is

$$L_{\text{Huber}}(\theta) = \sqrt{2 \cdot \frac{1}{T} \sum_{j=1}^T \rho_\delta(e_j(\theta))}.$$

This matters because sparse clinical trajectories often contain outlying jumps or measurement inconsistencies. A model that only succeeds under pure least-squares may be too fragile for later SBI.

3.7 Observation-Shift Extension

Early residual plots revealed cases where the mechanistic shape looked plausible but the later observations appeared shifted upward or downward. A multiplicative observation changepoint was introduced:

$$\log \hat{r}_j^{\text{shift}} = \log \hat{r}(t_j; \theta) + b \mathbf{1}_{j \geq k},$$

where k is a breakpoint index and $\exp(b)$ is the multiplicative shift factor. This is not a biological effect model. It is a measurement-process diagnostic layer.

3.8 Nested Resistant-Simeoni Model

The first two-compartment sensitive/resistant extension was not properly nested in the original Simeoni model and often degraded fits. The key modeling correction was the *nested resistant-Simeoni* extension, which preserves the original sensitive damage chain and adds a resistant compartment:

$$x(t) = (s_1(t), s_2(t), s_3(t), s_4(t), r(t)),$$

with total burden

$$V(t) = s_1 + s_2 + s_3 + s_4 + r.$$

The parameters are

$$\theta_{\text{NR}} = (\lambda_{0,s}, \lambda_1, \psi_g, k_{\text{kill},s}, k_{\text{tr}}, v_0, f_{r0}, m_r, q_r, k_{sr}).$$

The resistant parameters are tied to the sensitive ones through

$$\lambda_{0,r} = m_r \lambda_{0,s}, \quad k_{\text{kill},r} = q_r k_{\text{kill},s}.$$

The ODE becomes

$$\begin{aligned} \frac{ds_1}{dt} &= g_s(s_1) - k_{\text{kill},s}C(t)s_1 - k_{sr}s_1, \\ \frac{ds_2}{dt} &= k_{\text{kill},s}C(t)s_1 - k_{\text{tr}}s_2, \\ \frac{ds_3}{dt} &= k_{\text{tr}}(s_2 - s_3), \\ \frac{ds_4}{dt} &= k_{\text{tr}}(s_3 - s_4), \\ \frac{dr}{dt} &= g_r(r) - k_{\text{kill},r}C(t)r + k_{sr}s_1. \end{aligned}$$

This model approximately reduces to the plain Simeoni model when

$$f_{r0} \approx 0, \quad k_{sr} \approx 0.$$

3.9 Notebook 00 Outputs

On the first showcased patient, DATASET_263 / 203705005, using the patient-specific dose schedule reduced differential-evolution log-RMSE from roughly 0.273 under a generic q2w schedule to roughly 0.202 under the actual dose schedule. This established a key principle for the rest of the project: even simple tumor models benefit materially from correct dosing context.

For the selected retrospective 20-patient batch, the main results were:

Table 2: Notebook 00 retrospective model comparison over the 20-patient batch.

Model	Median RMSE	Mean RMSE	Min	Max
FinalSelectedConservative	0.151	0.181	0.036	0.530
NestedResistantSimeoni_plus_posthoc_shift	0.151	0.176	0.035	0.485
NestedResistantSimeoni	0.178	0.200	0.036	0.500
Simeoni_plus_joint_observation_shift	0.238	0.271	0.058	0.612
Simeoni	0.290	0.290	0.084	0.612

The final conservative model counts were:

- NestedResistantSimeoni_plus_posthoc_shift: 7 patients,
- NestedResistantSimeoni: 6 patients,

- Simeoni: 6 patients,
- Simeoni_plus_joint_observation_shift: 1 patient.

The median RMSE improved from

$$0.2898 \quad (\text{plain Simeoni})$$

to

$$0.2405 \quad (\text{best simple existing model})$$

and then to

$$0.1508 \quad (\text{final conservative selected model}).$$

Interpretation. Notebook 00 delivered a strong retrospective modeling result and a strong warning at the same time. Richer dynamics and observation layers clearly improved fit. But this was *in-sample* evidence only. The next notebook had to ask whether these apparent improvements generalized to future scans or simply overfit sparse trajectories.

4 Notebook 01: Hold-Out Model Validation

Notebook 01 is where the project first stress-tested the optimism created by Notebook 00. The question changed from

Can the model explain an entire observed trajectory?

to

Can the model trained on early scans predict later scans?

4.1 Train/Test Split Logic

For each of the same 20 patients, the trajectory was split chronologically. Roughly the first

75%

of scans formed the training segment and the last

25%

formed the hold-out segment. Models were fit only on the training prefix and evaluated on future scans.

This is critical because a sparse longitudinal fit can look impressive if it is allowed to see the entire future trajectory during optimization.

4.2 Hold-Out Objective and Complexity Penalty

Training still used log-RMSE-like objectives, but evaluation focused on future prediction error. Notebook 01 also tracked BIC-like complexity-aware comparison. In generic form,

$$\text{BIC} = n \log \left(\frac{\text{RSS}}{n} \right) + k \log n,$$

where k is the number of fitted parameters. The notebook compared:

- rule-based conservative selection,
- BIC-based selection,
- per-model train and test RMSE summaries.

4.3 Notebook 01 Outputs

The key numerical summary is shown in Table ??.

Table 3: Notebook 01 hold-out validation summary. Training fit improved with flexibility, but future test error remained high.

Model	Median train RMSE	Median test RMSE	Mean test RMSE	Median test RMSE
NestedResistantSimeoni	0.172	0.488	0.646	
Simeoni_plus_joint_observation_shift	0.197	0.495	0.591	
NestedResistantSimeoni_plus_posthoc_shift	0.154	0.498	0.613	
Simeoni	0.216	0.513	0.602	

The conservative rule-based selection counts were:

- Simeoni: 9,
- NestedResistantSimeoni: 6,
- NestedResistantSimeoni_plus_posthoc_shift: 4,
- Simeoni_plus_joint_observation_shift: 1.

The BIC selection counts were:

- Simeoni: 14,
- Simeoni_plus_joint_observation_shift: 3,
- NestedResistantSimeoni: 2,
- NestedResistantSimeoni_plus_posthoc_shift: 1.

Most importantly, the median test RMSE of the rule-selected model was

0.4246,

while the median test RMSE of the BIC-selected model was

0.5257.

4.4 What Notebook 01 Changed Scientifically

Notebook 01 showed that the retrospective success of Notebook 00 did not transfer cleanly to future scans. Mechanistic flexibility lowered training error substantially, but median test RMSEs still clustered near

0.49–0.51,

which is much worse than the training fit quality.

This is the first major identifiability warning:

a mechanistic model can interpolate early scans well while still failing as a forecasting model

because the data do not constrain the hidden parameters sharply enough.

That finding motivated Notebook 02, where the project stopped treating “future prediction” as a single split problem and switched to repeated rolling-origin forecasting.

5 Notebook 02: Forecasting Baselines and Rolling-Origin Validation

Notebook 02 reframed the problem in a more practical way. Instead of one train/test split per patient, it asked:

Given scans $1, \dots, j$, how well can we predict scan $j + 1$?

5.1 Rolling-Origin Forecasting Setup

At each forecast origin j , the model sees a prefix

$$\{(t_1, r_1), \dots, (t_j, r_j)\}$$

and predicts the next relative tumor size. The target is therefore a one-step predictive distribution or point prediction for

$$r_{j+1} \mid r_{1:j}, u_{1:j}.$$

The loss on each origin is based on the log error

$$\varepsilon_{j+1} = \log r_{j+1} - \log \hat{r}_{j+1}.$$

Notebook 02 reports

- median absolute log error,
- mean absolute log error,
- one-step RMSE.

5.2 Empirical Baselines

Three empirical baselines are especially important:

LOCF. Last observation carried forward predicts

$$\hat{r}_{j+1} = r_j.$$

This is a persistence baseline.

RecentLogSlope. Use the most recent local slope on log scale,

$$\widehat{\log r}_{j+1} = \log r_j + \frac{\log r_j - \log r_{j-1}}{t_j - t_{j-1}}(t_{j+1} - t_j).$$

LogLinear. Fit a least-squares line to the available prefix in log space,

$$\log r(t) \approx \beta_0 + \beta_1 t,$$

and extrapolate to the next scan time.

These baselines are scientifically useful because they set the minimum forecasting standard that a mechanistic simulator must beat before being trusted for SBI.

5.3 Mechanistic Competitors

Notebook 02 compares the empirical baselines against:

- Simeoni,
- Simeoni_plus_joint_observation_shift,
- NestedResistantSimeoni,
- NestedResistantSimeoni_plus_train_shift.

The “train shift” variant only uses information available up to the forecast origin, preventing direct look-ahead leakage.

5.4 Notebook 02 Outputs

The main one-step forecasting results are in Table ??.

Table 4: Notebook 02 rolling-origin one-step forecasting on the 20-patient cohort.

Model	N origins	Median log err	Mean log err	RMSE
LOCF	134	0.046	0.102	0.185
RecentLogSlope	134	0.066	0.127	0.210
LogLinear	134	0.243	0.295	0.376
NestedResistantSimeoni_plus_train_shift	134	0.234	0.313	0.404
NestedResistantSimeoni	134	0.239	0.342	0.462
Simeoni_plus_joint_observation_shift	134	0.312	0.411	0.542
Simeoni	134	0.320	0.442	0.582

Selection strategies were also compared:

Table 5: Notebook 02 rolling-origin selection strategy comparison.

Strategy	Count	Median log err	RMSE
oracle_per_origin	134	0.026	0.105
global_best_LOCF	134	0.046	0.185
adaptive_past_patient_errors	134	0.049	0.204

The per-patient best-model counts were:

- LOCF: 12 patients,
- RecentLogSlope: 7 patients,
- Simeoni: 1 patient.

5.5 Interpretation

Notebook 02 changes the direction of the whole project. The lesson is not just that mechanistic models are imperfect. The lesson is that on sparse real clinical tumor trajectories, a simple persistence forecast is extremely hard to beat:

if a mechanistic model cannot beat LOCF in rolling-origin prediction, retrospective fit alone is not enough ev

This is why Notebook 03 pivots to probabilistic forecasting and uncertainty calibration. Once point forecasting is difficult, the next scientifically meaningful question becomes: can we at least produce calibrated predictive intervals?

6 Notebook 03: Probabilistic Forecasting and Calibration

Notebook 03 makes a conceptual move rather than only a numerical one. It treats deterministic one-step predictions as mean structures and adds a residual uncertainty model on top.

6.1 From Point Forecasts to Predictive Distributions

Suppose a deterministic forecaster returns $\hat{r}_{j+1}$. Notebook 03 models the next observation in log space as

$$\log r_{j+1} = \log \hat{r}_{j+1} + \epsilon_{j+1}.$$

Residuals from rolling-origin backtesting are then used to estimate the distribution of ϵ .

Two main strategies are used.

Gaussian calibration. Assume

$$\epsilon \sim \mathcal{N}(\mu, \sigma^2),$$

usually with μ close to zero after calibration. Then a predictive interval for level $1 - \alpha$ is

$$\log \hat{r}_{j+1} \pm z_{1-\alpha/2} \sigma.$$

The Gaussian negative log likelihood is

$$\text{NLL} = \frac{1}{2} \log(2\pi\sigma^2) + \frac{(\epsilon - \mu)^2}{2\sigma^2}.$$

Empirical-quantile calibration. Instead of imposing Gaussian tails, estimate residual quantiles directly. If $q_{\alpha/2}$ and $q_{1-\alpha/2}$ are empirical residual quantiles, the predictive interval becomes

$$[\log \hat{r}_{j+1} + q_{\alpha/2}, \log \hat{r}_{j+1} + q_{1-\alpha/2}].$$

This is especially attractive when residuals are skewed or heavy tailed.

6.2 Coverage, Width, and PIT Diagnostics

Notebook 03 formalizes several metrics that remain central later:

- coverage at nominal levels such as 50%, 80%, 90%, and 95%;
- interval width, which controls sharpness;
- Gaussian NLL, which rewards both accuracy and calibrated uncertainty;
- probability integral transform (PIT) diagnostics for calibration shape.

If F_j is the predictive CDF for observation y_j , then

$$\text{PIT}_j = F_j(y_j)$$

should be approximately uniform on $[0, 1]$ under perfect calibration.

6.3 Why Notebook 03 Matters

Notebook 03 is theoretically important because it shifts the goal from

best deterministic forecast

to

best calibrated predictive distribution.

That shift becomes even more important once later notebooks show that raw mechanistic parameters are weakly identifiable. Even when parameters are poorly recovered, useful predictive uncertainty may still be achievable.

Notebook 04 operationalizes these ideas at full cohort scale and under leakage-free evaluation.

7 Notebook 04: Cohort-Scale Leakage-Free Baselines and SBI Readiness

Notebook 04 scales the rolling-origin and calibration framework from a 20-patient exploratory cohort to the canonical cross-study dataset.

7.1 Cohort Definition

The forecasting cohort applies stricter filters:

at least 6 scans, duration at least 90 days, minimum training scans = 4.

This produced:

- 5,862 patients in the overall feature table,
- 1,923 patients in the forecasting cohort,
- 7,710 deterministic one-step predictions,
- 23,130 long-format prediction rows across multiple models and calibration views.

7.2 Leakage-Free Calibration Pools

Residual pools are formed without using the target patient’s own future prediction error. Notebook 04 tests several calibration scopes:

- `global_lopo`: global leave-one-patient-out,
- `gapbin_global_lopo`: global but stratified by scan-gap bin,
- `study_lopo`: leave-one-patient-out within study,
- `drug_lopo`: leave-one-patient-out within drug signature,
- `study_drug_lopo`: leave-one-patient-out within study-drug grouping.

The goal is not only calibration accuracy but also realism: uncertainty should adapt to the context in which the forecast is made.

7.3 Deterministic Baselines at Scale

The cohort-scale deterministic results are shown in Table ??.

Table 6: Notebook 04 deterministic empirical baselines over the cohort-scale rolling-origin task.

Model	Predictions	Patients	Median log err	RMSE
LOCF	7,710	1,923	0.068	0.302
RecentLogSlope	7,710	1,923	0.105	0.359
LogLinear	7,710	1,923	0.224	0.438

LOCF remains the best deterministic model. This is an important continuity check with Notebook 02: the dominance of persistence was not a small-cohort fluke.

7.4 Probabilistic Calibration Outputs

Notebook 04 identifies two especially relevant calibrated baselines.

Best Gaussian calibration. The best Gaussian calibration uses LOCF with `gapbin_global_lopo`. Its key values are:

$$\begin{aligned}\text{RMSE} &= 0.3015, \\ \text{mean Gaussian NLL} &= 0.2032, \\ \text{coverage}_{80} &= 0.9043, \\ \text{coverage}_{90} &= 0.9333, \\ \text{coverage}_{95} &= 0.9514.\end{aligned}$$

The 95% interval is almost perfectly calibrated, while lower nominal intervals are conservative.

Best empirical-quantile calibration. The most accurate nominal coverage comes from RecentLogSlope with `global_lopo` empirical quantiles:

$$\begin{aligned}\text{coverage}_{50} &= 0.4997, \\ \text{coverage}_{80} &= 0.7997, \\ \text{coverage}_{90} &= 0.8999, \\ \text{coverage}_{95} &= 0.9499,\end{aligned}$$

with mean absolute coverage error essentially zero. This calibration is extremely well matched to nominal probabilities, though the underlying deterministic forecaster is weaker than LOCF.

Table ?? summarizes the two main baselines.

Table 7: Notebook 04 strongest leakage-free calibrated baselines.

Model	Scope	Interval	Cov.50	Cov.80	Cov.90	Cov.95	Width80	Mean cov.
LOCF	<code>gapbin_global_lopo</code>	Gaussian	0.781	0.904	0.933	0.951	0.654	0
RecentLogSlope	<code>global_lopo</code>	Emp. quantile	0.500	0.800	0.900	0.950	0.520	0.00

7.5 SBI Readiness by Study/Drug/Arm Stratum

Notebook 04 does more than score forecasting baselines. It introduces a concrete SBI-readiness layer. The rationale is:

Do not train SBI across all trials indiscriminately.

Instead, select coherent study/drug/arm strata with adequate scan depth, dose availability, manageable baseline error, and interpretable mechanism hints.

The readiness counts were:

- green: 8 strata,
- yellow: 8 strata,
- red: 17 strata.

The top recommended starter strata are shown in Table ??.

Table 8: Top SBI starter strata from Notebook 04.

Study	Arm	Family hint	Patients	Median scans	Baseline RMSE	Cov.90
CEDIRANIB_HORIZONIII	FOLFIRI + bevacizumab 5mg/kg	anti-angiogenic	268	6	0.297	0.931
DATASET_264	FOLFOX alone	cytotoxic TGI	190	7	0.361	0.891
EFC10262	Placebo + FOLFIRI	cytotoxic TGI	190	8	0.358	0.932
DATASET_264	Panitumumab + FOLFOX	cytotoxic TGI	188	8	0.374	0.867
DATASET_263	Panitumumab + FOLFIRI	cytotoxic TGI	152	7	0.306	0.915
A6181122	UNKNOWN_ARM	cytotoxic TGI	127	8	0.331	0.911
EFC5505	PLACEBO	cytotoxic TGI	120	7	0.498	0.877
DATASET_263	FOLFIRI alone	cytotoxic TGI	104	7	0.335	0.907

7.6 Why Notebook 04 Is a Turning Point

Notebook 04 formalizes three project-level rules:

1. the official predictive baseline is leakage-free calibrated empirical forecasting, not retrospective ODE fit;
2. SBI should begin in coherent green/yellow strata, not across the entire pooled dataset;
3. SBI should be evaluated by posterior predictive calibration, NLL, interval width, and long-horizon behavior.

The top green stratum was

CEDIRANIB_HORIZONIII / FOLFOX + bevacizumab 5 mg/kg,

which becomes the anchor of Notebook 05.

8 Literature and Model Catalogue

The literature/model catalogue runs in parallel with the forecasting notebooks. Its role is to stop the project from adding compartments blindly just because a richer model helped one patient.

8.1 Mechanism-to-Model Routing

The catalogue maps drug classes and observed trajectory patterns to model families:

- cytotoxic chemotherapy → Simeoni TGI or nested resistant-Simeoni,
- targeted therapy → Emax or resistance-aware TGI,
- anti-angiogenic therapy → carrying-capacity / vascular-support models,
- immune checkpoint therapy → delayed immune-response models,
- combination therapy → multi-input interaction TGI,
- lesion-rich data → lesion-level hierarchical models.

8.2 Why Anti-Angiogenic Therapy Needs a Different Latent Story

The catalogue explicitly warns against representing anti-VEGF-like treatment as direct cytotoxic kill alone. A reduced carrying-capacity model uses tumor burden $V(t)$ and vascular support $K(t)$:

$$\begin{aligned}\frac{dV}{dt} &= \rho V \left(1 - \frac{V}{K}\right), \\ \frac{dK}{dt} &= k_{K,\text{recovery}}(K_0 - K) - k_{\text{anti}}C(t)K.\end{aligned}$$

This is the conceptual starting point for the anti-angiogenic simulator later used in Notebooks 05C and 06.

8.3 Strategic Use of the Catalogue

The catalogue changes model development from

“what extra compartment should we add next?”

to

“what latent structure is justified by drug mechanism, data granularity, and identifiability?”

That is why Notebook 05 does not simply export the original all-purpose Simeoni model. It exports simulator families aligned with specific strata.

9 Notebook 05: Selected-Stratum Simulator Design and Synthetic SBI Preparation

Notebook 05 is where the project transitions from real-data forecasting diagnostics to synthetic SBI design.

9.1 Selected Target Stratum

The chosen first stratum is

CEDIRANIB_HORIZONIII / FOLFOX + bevacizumab 5 mg/kg.

Its readiness summary from Notebook 04 is:

- 268 real patients,
- median 6 scans,
- median duration 362 days,
- median gap 64 days,
- full dose availability,
- baseline RMSE 0.297,
- baseline 90% coverage 0.931.

The response-class fractions in the real cohort are:

response or shrinkage = 0.459,
response then regrowth = 0.325,
stable-like = 0.198,
progression or growth = 0.019.

9.2 Synthetic Simulator Philosophy

Notebook 05 does not simulate artificial evenly spaced trajectories from generic priors alone. It reuses real patient templates:

- real scan schedules,
- real dose schedules,
- real trajectory lengths,
- real baseline scales,
- clinically coherent study/drug/arm metadata.

This is crucial. For SBI, the observation design itself is part of the inverse problem.

9.3 Simulator Families

The first 05 export used two simulator families.

Simeoni family. This retains the cytotoxic transit-damage model and serves as a mechanistic baseline family.

Angio family. For anti-angiogenic behavior, a reduced tumor-plus-carrying-capacity model is used:

$$\begin{aligned}\frac{dV}{dt} &= \rho V \left(1 - \frac{V}{K}\right), \\ \frac{dK}{dt} &= k_{K,\text{recovery}}(K_0 - K) - k_{\text{anti}}C(t)K.\end{aligned}$$

The initial capacity is parameterized as

$$K(0) = K_0 = K0_ratio \cdot V(0),$$

and the core angio parameter vector becomes

$$\theta_{\text{angio}} = (\rho, k_{\text{anti}}, k_{K,\text{recovery}}, K0_ratio, v_0).$$

This reduced model is intentionally low-dimensional. It is designed to test whether anti-angiogenic local dynamics can be inferred at all from sparse relative SLD trajectories.

9.4 Initial 05 Export

The first synthetic export contained:

- 2,000 total simulations,
- 1,000 Simeoni simulations,
- 1,000 angio simulations,
- array shape $\mathbf{x}_{\text{obs}} \in \mathbb{R}^{2000 \times 14 \times 3}$,
- parameter array shape $\theta \in \mathbb{R}^{2000 \times 10}$ in the mixed-family export.

The mixed theta names were

$$(\theta_{K0_ratio}, \theta_{k_{K_recovery}}, \theta_{k_{\text{anti}}}, \theta_{k_{\text{cyt}}}, \theta_{k_{\text{tr}}}, \theta_{\lambda 0}, \theta_{\lambda 1}, \theta_{\psi_g}, \theta_{\rho}, \theta_{v_0}),$$

with a simulator ID indicating whether the sample came from the Simeoni or angio family.

9.5 Why 05 Needed Further Tuning

The first prior-predictive cohort did not match the real stratum closely enough. In the target CEDIRANIB stratum, the synthetic response-class fractions were too progression-heavy:

- angio progression/growth: 0.422,
- Simeoni progression/growth: 0.557,
- real progression/growth: only 0.0187.

That mismatch is not a cosmetic issue. If the synthetic generator produces the wrong local dynamics, an NPE trained on it will learn the wrong inverse mapping.

10 Notebook 05B and 05C: Tuned Multistratum Synthetic Exports

The 05B and 05C updates are where the simulator preparation becomes mature enough for Notebook 06.

10.1 05B: Prior Tuning and Multistratum Export

Notebook 05B expands from one target stratum to four selected strata and tunes class quotas so the synthetic response-pattern mixture better resembles the real data.

The four strata are:

1. CEDIRANIB_HORIZONIII / FOLFOX + bevacizumab 5 mg/kg,
2. DATASET_264 / FOLFOX alone,
3. EFC10262 / Placebo + FOLFIRI,
4. DATASET_264 / Panitumumab + FOLFOX.

The tuned export contains

4,800

synthetic trajectories total:

2,400 Simeoni and 2,400 angio.

Each study-arm-stratum contributes 600 simulations per simulator family.

The grouped split is by real template rather than by individual simulation row, which is essential to prevent template leakage between train and test.

10.2 05C: Local-Dynamics Matching

Notebook 05C refines the export again, this time emphasizing local one-step dynamics and real-vs-synthetic residual matching. The final 05C exports are:

$$\mathbf{x}_{\text{obs}} \in \mathbb{R}^{2400 \times 25 \times 3},$$
$$x_{\text{cov}} \in \mathbb{R}^{2400 \times 8},$$

with covariates

(log baseline SLD, duration_years, $n_scans/25$, dose_confounder, 4 stratum one-hots).

The train/validation/test split sizes for the angio export are:

1948, 218, 234.

10.3 05C Real-vs-Synthetic Output Matching

The response summaries in Table ?? show how much closer the tuned synthetic cohorts came to the real strata.

Table 9: Selected 05C real-vs-synthetic response summaries.

Source / stratum	Simulator	Median final rel. SLD	Median minimum rel. SLD	Shrink frac	Regrowth frac
CEDIRANIB real	real	0.629	0.521	0.459	0.323
CEDIRANIB 05C	angio	0.459	0.392	0.453	0.323
CEDIRANIB 05C	simeoni	0.311	0.256	0.453	0.323
FOLFOX-alone real	real	0.619	0.404	0.389	0.458
FOLFOX-alone 05C	angio	0.525	0.381	0.390	0.458
FOLFOX-alone 05C	simeoni	0.286	0.212	0.390	0.458

The local residual behavior is even more informative. For the target CEDIRANIB stratum:

real LOCF RMSE = 0.231, angio synthetic LOCF RMSE = 0.176, simeoni synthetic LOCF RMSE =

Thus the angio simulator is much closer than the Simeoni family to the local dynamics of the anti-angiogenic real cohort.

Table ?? summarizes the key values.

Table 10: 05C local-dynamics matching via LOCF residual summaries.

Source / stratum	Simulator	Count	Median log err	RMSE	q_{95}
CEDIRANIB real	real	772	0.064	0.231	0.363
CEDIRANIB synthetic	syn- angio	1,670	0.112	0.176	0.333
CEDIRANIB synthetic	syn- simeoni	1,661	0.250	0.486	0.555
EFC10262 real	real	850	0.043	0.384	0.338
EFC10262 synthetic	angio	2,306	0.077	0.144	0.262
EFC10262 synthetic	simeoni	2,162	0.131	0.307	0.349

10.4 Why 05C Becomes the Official Notebook 06 Input

Notebook 05C is the correct launch point for NPE because it satisfies three conditions:

1. grouped template splits avoid train/test leakage,
2. local dynamics are substantially closer to real data than the naive export,
3. the angio simulator matches the anti-angiogenic target stratum far better than the plain Simeoni family.

That is why Notebook 06 uses the 05C angio-only export rather than the earlier mixed-family array.

11 Notebook 06: First Neural Posterior Estimation on 05C Angio

Notebook 06 asks the first truly SBI-specific question:

Can a learned posterior recover simulator parameters from synthetic tumor trajectories?

11.1 Input Representation

For each simulation, the padded observation sequence is

$$\mathbf{x}_{\text{obs}} = [\tilde{t}, \log r, m] \in \mathbb{R}^{T_{\text{max}} \times 3},$$

where $\tilde{t}$ is normalized time, $\log r$ is log-relative tumor burden, and m is a mask for variable-length trajectories.

The covariates are

$$x_{\text{cov}} \in \mathbb{R}^8,$$

with baseline scale, duration, scan count, dose-confounding flag, and one-hot stratum indicators.

The 05C angio target parameters are

$$\theta = (\rho, k_{\text{anti}}, k_{K,\text{recovery}}, K0_ratio, v_0).$$

Notebook 06 standardizes them on log scale:

$$z = \frac{\log \theta - \mu_{\log \theta}}{\sigma_{\log \theta}}.$$

11.2 Posterior Families and Baselines

Notebook 06 compares:

- a prior Gaussian baseline,
- Ridge residual bootstrap,
- Random Forest residual bootstrap,
- summary-feature MLP regression,
- summary-feature MDNs,
- GRU-based MDNs.

The core posterior density family is a mixture density network:

$$q_{\phi}(z | x) = \sum_{k=1}^K \pi_k(x) \mathcal{N}(z; \mu_k(x), \text{diag}(\sigma_k^2(x))).$$

The GRU variant uses the sequential trajectory directly; the summary models compress the trajectory into engineered features first.

11.3 Train/Validation/Test Split

The 05C angio export used here contains:

- $N = 2400$ total simulations,
- $T_{\text{max}} = 25$,
- train/val/test sizes 1948/218/234,
- 757 unique real templates,
- four strata.

Because the split is template-grouped, the task is harder and more realistic than a random row split.

11.4 Notebook 06 Aggregate Results

The selected best model is

`gru_mdn_K5_H128`.

Its aggregate held-out metrics are

$$\begin{aligned} \text{mean RMSE}_z &= 0.858, \\ \text{mean MAE}_z &= 0.727, \\ \text{mean RMSE}_{\log \theta} &= 0.883, \\ \text{mean abs. relative error} &= 0.981, \\ \text{mean corr}_z &= 0.454, \\ \text{mean coverage}_{90} &= 0.915, \\ \text{mean posterior std}_z &= 0.866. \end{aligned}$$

This is better than the prior and slightly better than the Ridge / Random Forest baselines, but the margin is not large.

Parameter-level performance is shown in Table ??.

Table 11: Notebook 06 best-model parameter recovery on held-out 05C angio simulations.

Parameter	RMSE _z	Corr _z	Mean rel. err	Cov.90	Posterior std _z
ρ	0.905	0.450	0.928	0.906	0.913
k_{anti}	0.928	0.384	1.506	0.906	0.901
$k_{K,\text{recovery}}$	0.908	0.377	1.315	0.944	0.939
$K0_ratio$	0.952	0.265	0.386	0.915	0.950
v_0	0.598	0.792	0.770	0.902	0.625

11.5 Interpretation of Notebook 06

Only v_0 is recovered moderately well, with correlation near

$$0.79.$$

This is not surprising because baseline tumor scale is directly represented in the covariates. The harder mechanistic parameters

$$\rho, \quad k_{\text{anti}}, \quad k_{K,\text{recovery}}, \quad K0_ratio$$

are recovered only weakly. Correlations are low to moderate and errors remain substantial.

This result is extremely important:

even under simulator-matched synthetic data, the angio parameters are only weakly recoverable

Therefore the main problem is not just neural architecture quality. It is the information content of sparse relative SLD trajectories.

12 Notebook 06B: Stabilized NPE and Identifiability Rescue Attempts

Notebook 06B is explicitly a diagnosis notebook. It asks whether poor performance is due to:

1. random-vs-grouped split shift,

2. unstable MDN training,
3. excessive target dimensionality,
4. bad parameterization of the latent dynamics,
5. or fundamental weak identifiability.

12.1 Stabilization Strategy

The patch replaces unstable MDN heads with simpler Gaussian posterior heads and tests reduced/composite targets. Instead of always predicting the full five raw angio parameters, Notebook 06B defines:

- `core2_growth_treatment`: $(\log \rho, \log k_{\text{anti}})$,
- `core3_logtheta`: $(\log \rho, \log k_{\text{anti}}, \log K0_ratio)$,
- `composite3`: growth, treatment-vs-growth, recovery-vs-treatment, and capacity coordinates.

The composite representation is meant to test whether the data identify *relative dynamical balances* better than raw rate constants.

12.2 Decision Rule

The patch compares each stabilized model to a prior baseline on the same target. If the relative RMSE improvement over the prior is small, the model is still weakly informative.

In general form, define

$$G = \frac{\text{RMSE}_{\text{prior}} - \text{RMSE}_{\text{model}}}{\text{RMSE}_{\text{prior}}}.$$

Notebook 06B interprets

$$5\% - 15\%$$

relative gain as a weak-identifiability regime rather than a strong inference success.

12.3 Notebook 06B Outputs

The strongest reduced/composite targets are summarized in Table ??.

Table 12: Notebook 06B stabilized Gaussian target-comparison summary.

Target	Dim	Best model	RMSE _z	Prior RMSE _z	Rel. gain	Cov.90
composite3	4	GRU Gaussian	0.900	1.006	10.5%	0.868
core2_growth_treatment	2	Summary Gaussian	0.907	1.007	9.95%	0.853
core3_logtheta	3	GRU Gaussian	0.924	1.001	7.72%	0.886

The best configuration is

$$\text{06B_gru_gaussian_composite_global_sigma},$$

with

$$\text{mean RMSE}_z = 0.900, \quad \text{relative improvement over prior} = 10.5\%.$$

However, that still falls into the weak-identifiability band defined by the notebook.

12.4 Interpretation of Notebook 06B

Notebook 06B is one of the most important notebooks in the whole sequence because it prevents the wrong conclusion. A naive reading of Notebook 06 might say:

“try a larger network”.

Notebook 06B shows why that is the wrong move. Reduced targets, composite targets, and stabilized Gaussian heads only produce modest improvements over the prior:

7.7% to 10.5%.

That is a scientific signal, not a training inconvenience. It means:

sparse relative SLD trajectories weakly identify composite growth/treatment balances, but not detailed raw a

The right next step is therefore posterior predictive validation, not parameter-regression escalation.

13 Integrated Scientific Interpretation

Across Notebooks 00–06B, the project asks increasingly difficult questions.

1. **Can the raw clinical files be made analysis-ready?** Yes. Ten studies were canonicalized, with 5,862 patients entering the feature table.
2. **Can mechanistic models fit retrospective trajectories?** Yes. On a selected 20-patient batch, median retrospective RMSE improved from 0.290 under plain Simeoni to 0.151 after conservative selection over nested resistance and observation shifts.
3. **Do those fits generalize to future scans?** Only weakly. Hold-out median test RMSEs remained around 0.49–0.51.
4. **Do mechanistic models beat simple forecasting baselines?** No. LOCF dominated rolling-origin point forecasting on both the 20-patient and 1,923-patient cohorts.
5. **Can useful predictive uncertainty be calibrated empirically?** Yes. Leakage-free Gaussian and empirical-quantile calibration produced strong coverage baselines.
6. **Can a clinically coherent SBI starter problem be isolated?** Yes. The CEDIRANIB anti-angiogenic stratum was selected as the first green starter stratum.
7. **Can synthetic training data be made structurally similar to the real local dynamics?** Partly yes. The 05C angio export is much closer than the naive or Simeoni-only synthetic alternatives.
8. **Can NPE recover raw simulator parameters on simulator-matched synthetic data?** Only weakly. The best GRU-MDN barely outperformed simpler baselines, and 06B showed only modest gains over the prior even for reduced targets.

The full project therefore delivers a valuable negative result:

the real difficulty is not just flexible modeling; it is weak identifiability from sparse tumor trajectories

That negative result is productive. It stops the project from over-interpreting neural posterior outputs and redirects it toward better scientific evaluation criteria.

14 What the Next Full Notebook Should Be

The next notebook should not be another architecture search. It should be a posterior predictive notebook:

`07_angio_posterior_predictive_checks_synthetic_then_real.ipynb`.

14.1 Goal

The target question is

Can posterior samples generate calibrated predictive bands over tumor trajectories?

not

Can we recover every mechanistic parameter precisely?

14.2 Posterior Predictive Workflow

For a fitted posterior $q_\phi(\theta | x)$, the posterior predictive procedure is

$$\theta^{(s)} \sim q_\phi(\theta | x),$$

$$\hat{y}_{1:T}^{(s)} \sim p(y_{1:T} | \theta^{(s)}, u),$$

followed by comparisons between the empirical trajectory and the posterior predictive ensemble.

14.3 Pass/Fail Metrics

The next notebook should report:

- predictive coverage at 80%, 90%, and 95%,
- interval width versus the empirical baselines from Notebook 04,
- posterior predictive NLL,
- calibration by stratum and scan-gap bin,
- robustness to sparse prefixes and local jumps,
- synthetic-first validation before any real-patient interpretation.

The standard for success is not “posterior samples look diverse.” The standard is:

posterior predictive performance at least matches calibrated empirical baselines

and ideally improves on them in clinically coherent strata.

15 Limitations

The current pipeline remains a deliberately simplified but rigorous pre-SBI program. Main limitations are:

- the exposure model is still an exponential forcing approximation, not full PK;
- SLD compresses lesion-level heterogeneity into a single scalar;
- observation shifts are phenomenological and may reflect multiple non-biological processes;

- synthetic angio dynamics are reduced effective dynamics, not a full vascular biology model;
- parameter inference is tested mainly on synthetic data and remains weak even there;
- the strongest real-data predictive baselines remain empirical rather than mechanistic.

These are not failures. They define the correct next stage of work.

16 Conclusion

The draft `notebook1_all_notebooks_updated.tex` has now been expanded into a full report that treats each notebook as a real modeling stage rather than a short summary bullet. The complete notebook sequence tells a coherent story:

1. clinical data can be canonicalized across heterogeneous oncology trials;
2. mechanistic TGI models can fit many retrospective trajectories well;
3. those fits do not guarantee good future prediction;
4. strong empirical rolling-origin baselines are hard to beat;
5. calibrated predictive uncertainty is feasible and should be the benchmark;
6. simulator choice must follow drug mechanism and data context;
7. synthetic SBI is feasible in carefully selected strata;
8. raw parameter recovery remains weak even under simulator-matched synthetic data.

The most important conclusion is therefore:

Sparse relative SLD trajectories are informative enough to support useful empirical forecasting and some coarse dynamical uncertainty quantification, but they do not strongly identify detailed mechanistic PK/PD parameters. The next scientifically meaningful milestone is posterior predictive calibration, not aggressive raw-parameter recovery.

That is a strong methodological result. It protects future SBI work from over-claiming and defines a realistic path toward a credible oncology posterior simulator.

A Notebook-to-Output Map

Notebook	Main purpose	Most important saved output or conclusion
00	Clinical audit and first mechanistic discovery	Ten studies canonicalized; retrospective 20-patient median RMSE improved to 0.151 after conservative model selection.
01	Hold-out validation	Future-scan median test RMSEs remained near 0.49–0.51, exposing overfit risk.
02	Rolling-origin forecasting	LOCF dominated with RMSE 0.185 on the 20-patient cohort.
03	Probabilistic calibration theory	Formalized Gaussian and empirical residual calibration, coverage, NLL, and PIT.
04	Cohort-scale leakage-free baselines	1,923-patient cohort; LOCF best deterministic baseline; near-perfect empirical interval coverage available.
05	Selected-stratum simulator design	CEDIRANIB anti-angiogenic starter stratum selected; first synthetic SBI arrays exported.
05B	Multistratum tuned export	Four-stratum, grouped-split synthetic export with 2,400 angio and 2,400 Simeoni simulations.
05C	Local-dynamics matching	Angio simulator brought much closer to real local residual behavior than Simeoni in the target anti-angiogenic stratum.
06	First angio NPE	Best GRU-MDN achieved mean $\text{RMSE}_z = 0.858$ but weak raw parameter recovery overall.
06B	Identifiability rescue attempt	Reduced/composite targets improved only 7.7%–10.5% over the prior, implying weak identifiability.

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